AC9M6N09 • Year 6 Mathematics

Mathematical Modelling with Rational Numbers and Percentages

Formulate, solve, validate and justify practical and financial decisions

We are learning to…

Students identify quantities, rates, percentages, constraints and assumptions, build a model, compare scenarios and communicate a recommendation supported by calculations and sensitivity checks.

Represent | **Reason** | **Calculate** | **Interpret**

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

use mathematical modelling to solve practical problems involving natural and rational numbers and percentages, including in financial contexts; formulate the problems, choosing operations and efficient calculation strategies, and using digital tools where appropriate; interpret and communicate solutions in terms of the situation, justifying the choices made

Compare two fundraising plans

plan	income	costs	net
A: 180 tickets at $12	**$2 160**	**$780 fixed + 15% fee**	**$1 056**
B: 150 tickets at $15	$2 250	$1 050 fixed + 8% fee	$1 020
decision	A higher net	but test attendance assumptions	

The model separates revenue, percentage fees and fixed costs. A recommendation should state assumptions about tickets sold and constraints such as venue capacity.

Use a modelling and sensitivity cycle

define decision → identify quantities/constraints → represent equations/table → calculate scenarios → validate and estimate → change one assumption → recommend and communicate

Digital spreadsheets can compare many scenarios, but formulas, units and assumptions must be explained and checked.

Build and annotate the model

Represent mathematical modelling with rational numbers and percentages and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Every number in the context used:** Include only relevant quantities and relationships.
- **Percentage fee calculated from the wrong base:** Identify whether it applies to revenue, cost or balance.
- **Cheapest option selected without constraints:** Check capacity, quality and requirements.
- **Spreadsheet output accepted without formula audit:** Explain and verify formulas.

Quick check

- Identify a percentage fee base.
- Write a net-income equation.
- Check a capacity constraint.
- Test one assumption.
- Communicate a recommendation.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.